Stereo Disparity and Relative Depth on OpenCV Aloe

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This experiment computes stereo disparity and a relative-depth proxy for OpenCV's Aloe image pair. Both images were reduced to 641×555 pixels and processed with StereoSGBM. The default result contains valid positive disparity at 70.5% of pixels. Regions on the plant and pot have larger median disparity than the background, as expected for nearer surfaces. The data used here do not include a verified metric baseline or focal length; the resulting depth values are therefore not distances in meters.

Aim and data

The aim was to produce disparity, inspect its failures, and demonstrate the inverse relationship between disparity and depth. The inputs were `aloeL.jpg` and `aloeR.jpg` from OpenCV's `samples/data` collection. Both originals are 1282×1110 pixels; the program scales each dimension by one half before matching. The provided pair is treated as rectified for horizontal matching [OpenCV]. Horizontal guide lines in the recorded input pair support visual inspection of the alignment.



Figure 1: The left and right Aloe inputs after resizing. Horizontal guide lines show the scanlines used for visual alignment checks.

Method

Stereo rectification maps corresponding epipolar lines to horizontal scanlines. The rectification paper by Fusiello et al. [Fusiello et al., 2000] explains this geometric step; the present code starts from the already aligned sample pair and performs matching. A grayscale StereoSGBM matcher used `minDisparity` $= 0$, `numDisparities` $= 128$, `blockSize` $=$

5, $P_1 = 200$, $P_2 = 800$, `uniquenessRatio` $= 10$, and a 100-pixel speckle window. OpenCV's signed fixed-point output was divided by 16 to obtain pixel disparity $d = u_L - u_R$. Nonpositive and nonfinite values were excluded.

For ideal horizontally rectified cameras with equal horizontal principal points, depth would satisfy $Z = fB/d$. The OpenCV Aloe example [OpenCV] *guesses* $f = 0.8w$, where $w = 641$ pixels after resizing. Following that illustration, the program used $f = 512.8$ px and $B = 1$ arbitrary baseline unit:

$$Z_{\text{proxy}} = \frac{512.8}{d}. \tag{1}$$

This proxy preserves the near/far ordering for positive disparities. Its numerical values cannot be converted to meters from the available inputs. The estimated focal length and baseline assumption are recorded in `summary.json`.

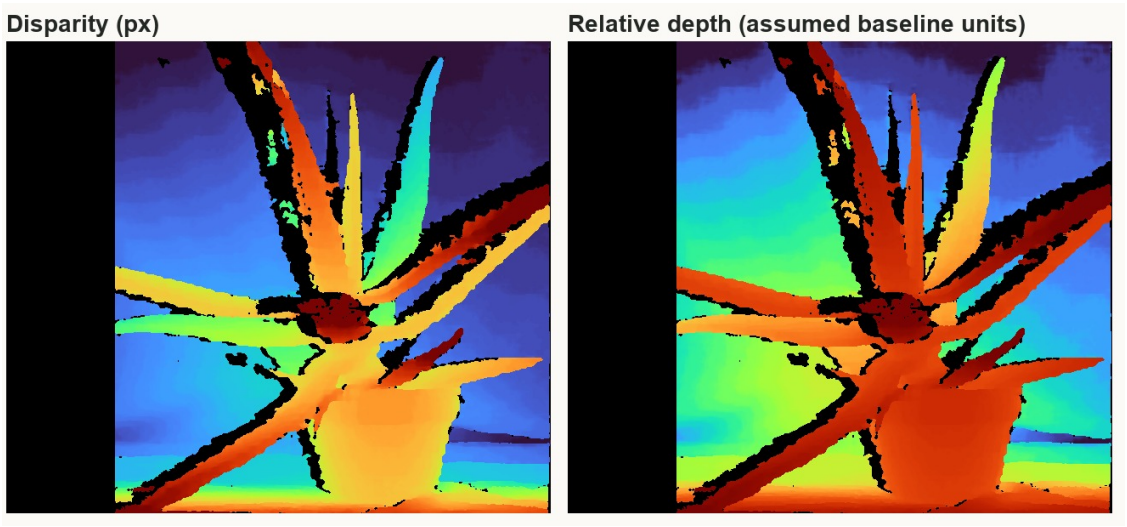


Figure 2: Computed disparity and illustrative relative depth. Black denotes excluded pixels. Colors are independently mapped over the 2nd–98th percentile range for visibility, so numeric comparisons must use the saved arrays rather than image colors.

Results

Of $641 \times 555 = 355{,}755$ pixels, 70.5% have valid positive disparity. Across those pixels, the 5th, median, and 95th disparity percentiles are 23.875, 32.063, and 65.250 px. The corresponding *distribution* of relative-depth values has 5th, median, and 95th percentiles of 7.859, 15.994, and 21.479 assumed baseline units. These distributions are reported separately; percentile positions reverse under the inverse mapping.

Three fixed regions were sampled: wallpaper background (B), central leaf/core (L), and pot (P). Their locations are marked in the retained figure asset.

Region	Valid (%)	Median d (px)	Median Z_{proxy} (baseline units)
Background (B)	52.0	25.69	19.96
Leaf/core (L)	78.1	58.00	8.84
Pot (P)	99.5	56.00	9.16

Table 1: Descriptive values over valid pixels inside the marked rectangles. They are not ground-truth accuracy measurements.

Increasing the matching block size from 5 to 9 changed the valid-pixel fraction from 70.5% to 70.8%. On pixels valid in both runs, the median absolute disparity difference was 0.062 px. This small aggregate change does not establish accuracy, especially at object edges and occlusions.

Interpretation and limits

The plant and pot show larger disparities than the patterned background, consistent with their closer position. Missing values occur near image borders, thin leaves, occlusions, and difficult texture. Repeated wallpaper motifs can also yield incorrect correspondences. No ground-truth disparity, measured baseline, or camera calibration was available for this pair. Consequently the experiment demonstrates relative depth only; a metric depth map would require calibrated stereo parameters and a matching rectification model.

Reproduction

The run used Python 3.9.23, NumPy 2.0.2, and OpenCV 4.12.0. From the repository root:

```
python3.9 lab0/code/stereo_depth.py -scale 0.5
```

The raw disparity, valid mask, relative-depth array, images, and run parameters are under `lab0/results/stereo/`. A second run set the block size to 9.

References

Andrea Fusiello, Emanuele Trucco, and Alessandro Verri. A compact algorithm for rectification of stereo pairs. *Machine Vision and Applications*, 12:16–22, 2000.

OpenCV. Stereo matching sample and aloe images. `https://github.com/opencv/opencv/tree/4.x/samples`. Official OpenCV source repository.